Fine-Tuning BLIP for Radiology Image Captioning

CSCI 4052U – Machine Learning II Final Project

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The Problem

Why Traditional CV Fails

Hand-engineered features (HOG, SIFT) cannot bridge the gap from raw pixels to fluent sentences.

The Challenge

Fusing visual understanding (anatomy, pathology) with domain-specific medical phrasing.

The Solution: Vision-Language Models (VLMs)

VLMs learn to generate text conditioned on image embeddings, creating a direct link between pixels and prose.

Neural Network Architecture

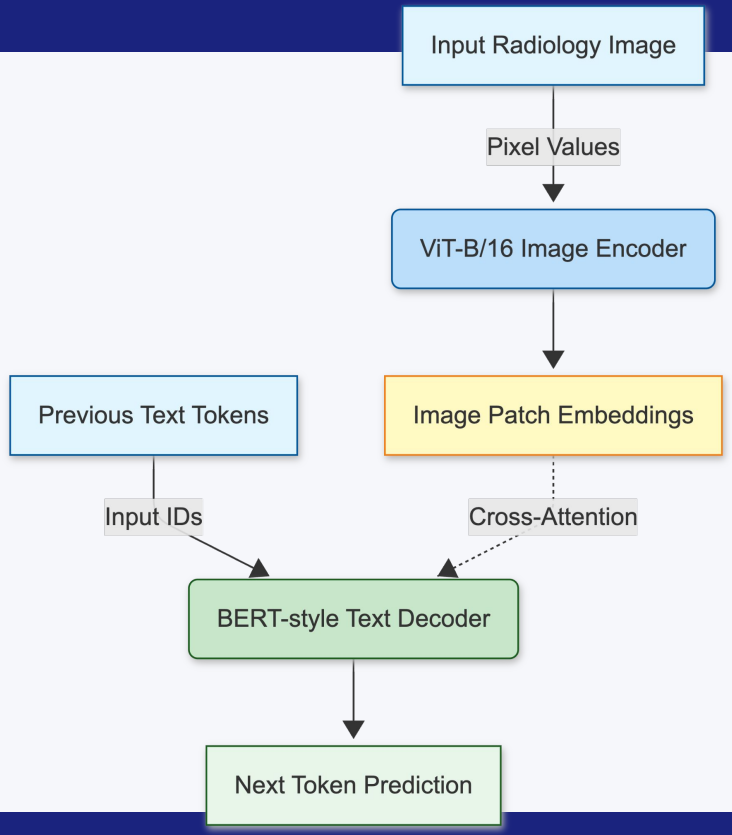
Model: BLIP
Bootstrapping Language-Image Pretraining

Vision Encoder: ViT-B/16
Turns image into a grid of patch embeddings.

Text Decoder: BERT-style
Generates captions autoregressively.

The Bridge
Cross-attention layers linking text tokens to image patches.

~224M Parameters
Optimized for fine-tuning on a single T4 GPU.



The Data: ROCO Dataset

Source & Processing

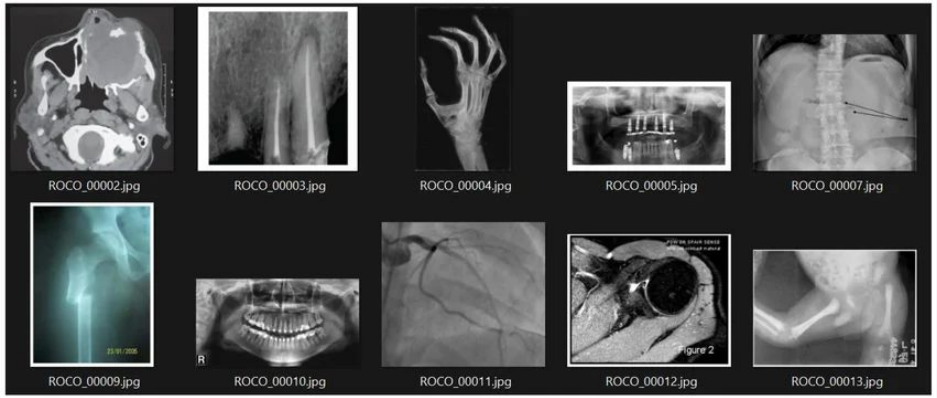
Scraped from open-access PubMed articles; auto-downloaded via HuggingFace.

Filtering

Captions restricted to 5–60 words to remove outliers and maintain quality.

Dataset Splits

8,000 training examples and 1,000 validation examples.



Fine-Tuning Pipeline

Data Encoding

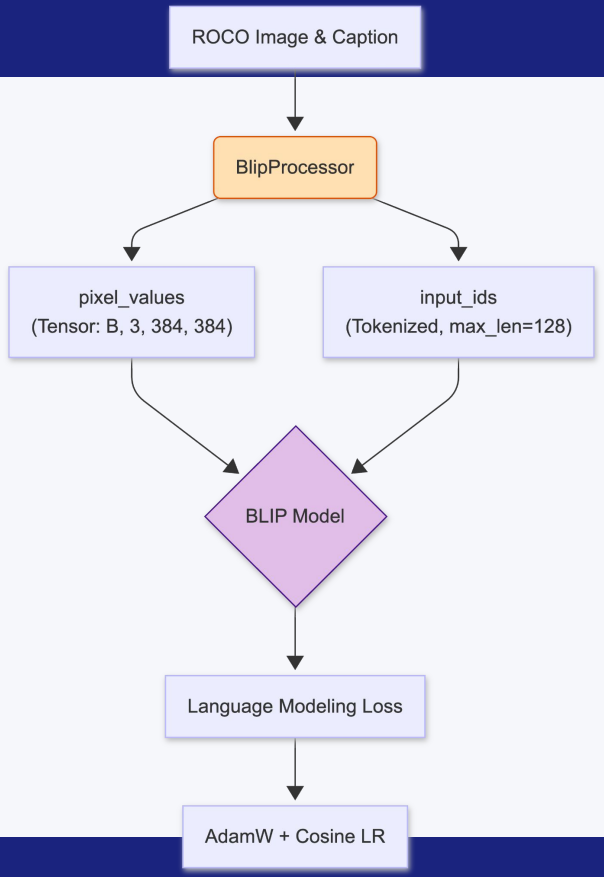
BlipProcessor normalizes images into tensors & tokenizes captions (max 128 tokens).

Strategy: Full Fine-Tuning

Preferred over Decoder-only to adapt the vision encoder to complex medical features.

Hyperparameters

AdamW, 5e-5 LR (Cosine), 16-bit precision (fp16) for efficient training.



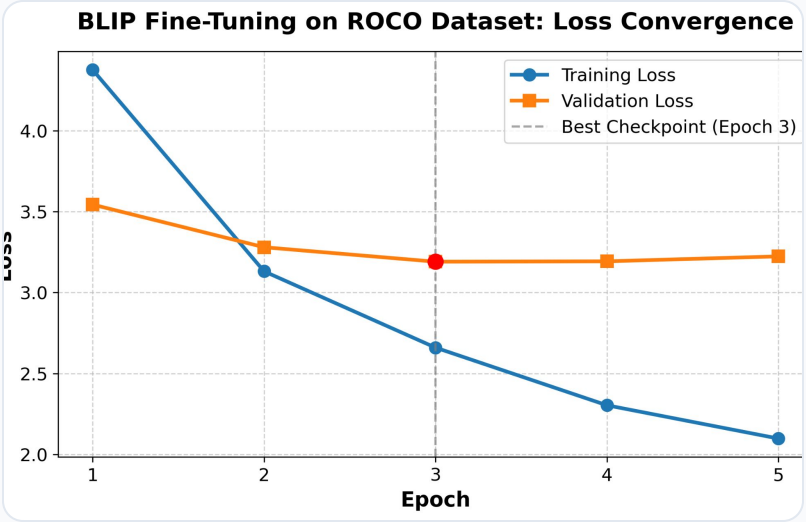
Training Run & Convergence

Hardware
Google Colab T4 GPU (16 GB VRAM)

Training Time
~52 minutes total wall-clock time

Convergence
Best validation loss: 3.1908 (Epoch 3)

Loss Convergence Curve



Quantitative Results

BLEU-4
Pretrained: 0.0011

0.0324

↑ +0.0313

METEOR
Pretrained: 0.0347

0.1644

↑ +0.1297

BERTScore-F1
Pretrained: 0.7041

0.7806

↑ +0.0765

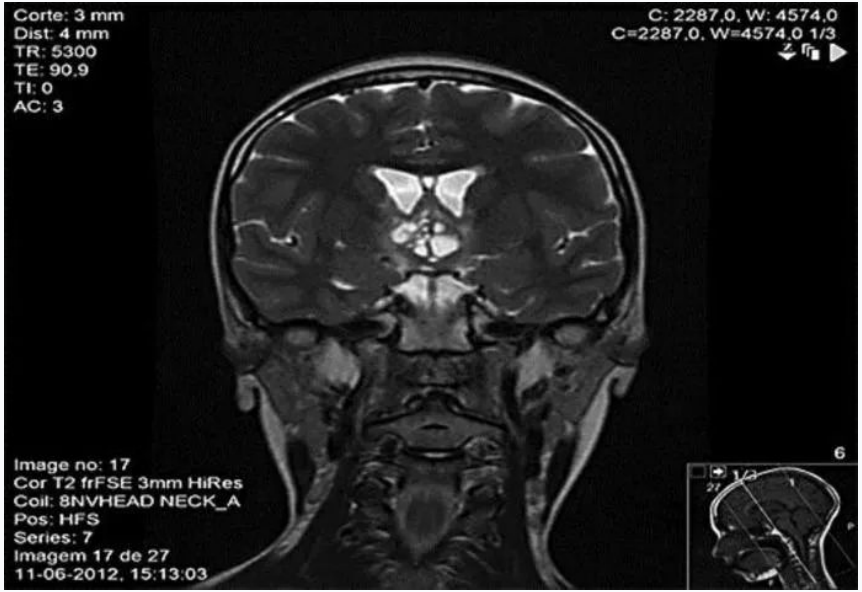
Key Takeaway
METEOR and BERTScore improvements highlight genuine domain vocabulary acquisition and semantic alignment.

Qualitative Results - Example 1

Comparative Captioning Details

- **Pretrained Model Output:** "an x - ray image of a man ' s chest and chest"
- **Fine-Tuned Model Output:** "postoperative x - ray of the patient."

Qualitative Results - Example 2



Comparative Captioning Details

- **Pretrained Model Output:** "a mri mri mri mri mri mri mri mri mri mri"
- **Fine-Tuned Model Output:** "coronal t2 - weighted magnetic resonance image of the head and neck showing a hyperintense lesion in the right parietal region."

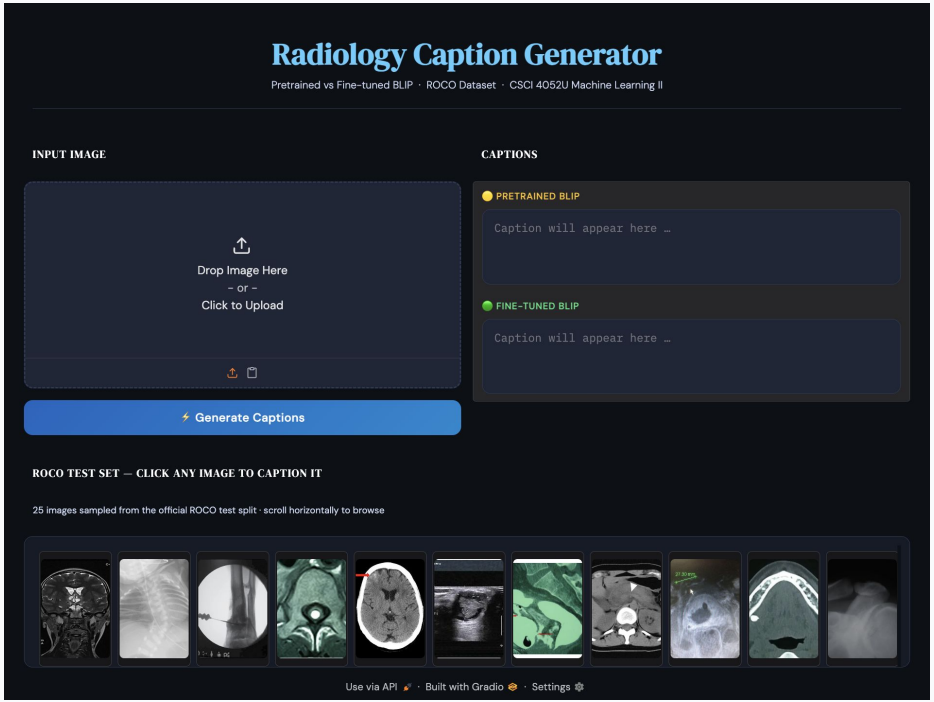
Deployment Pipeline

Environment
HuggingFace Spaces (Free CPU Tier)

Interface
Interactive Gradio Blocks UI

Integration
Loads weights via from_pretrained()

Key Features
Custom uploads & 25 ROCO test samples



Code Repository & Conclusion

GitHub Repository

https://github.com/abdulmuqit2004/image-caption-generator

Project Outcome

Successfully adapted general VLM to a highly specialized medical task.

Disclaimer

Educational research only; not for clinical use.

Thank You!